

ML FINAL

Math Topic
only

⇒ Decision tree ✓

⇒ Linear Regression ✓

⇒ Logistic Regression ✓

⇒ Adaboost ✓

⇒ ROC-AUC Curve ✓

⇒ Evaluation matrices

(precision, Recall, F1 score, Confusion matrices)

⇒ Loss Function

(MSE, MAE, Cross-Entropy)

⇒ Linear Regression

→ Direct Formula

→ According to slide ✓

$$\begin{aligned} & [2+, 2-] \\ & = 1 \end{aligned}$$

Machine Learning (CSE 457) CT-02 (CLO2)

Full Marks: 10

Time: 20 Minutes

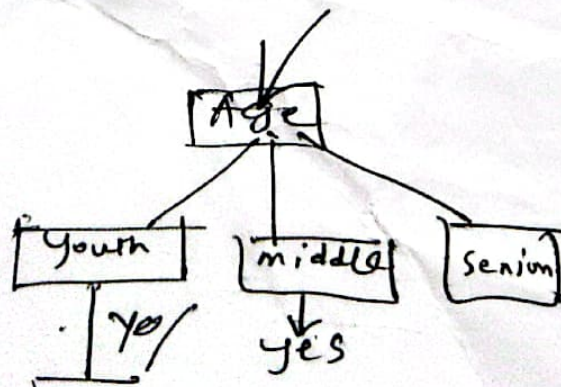
1. The following table consists of computer buying data from a local shop.

→ Target

	Age	Income	Student	Credit_rating	Buys_computer
1	youth →	high	no	fair	no
2	youth →	high	no	excellent	no
3	middle	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle	low	yes	excellent	yes
8	youth →	medium	no	fair	no
9	youth →	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth →	medium	yes	excellent	yes
12	middle	medium	no	excellent	yes
13	middle	high	yes	fair	yes
14	senior	medium	no	excellent	no

Calculate the entropy and information gain for each attribute and draw the first node of the tree using the decision tree algorithm.

(5) →



Topic 01 - Decision Tree

Attribute: Age

Values (Age) = {youth, middle, senior}

Step 01

Calculate whole entropy

$$S = [9+, 5-]$$

$$= (14)$$

9 $\rightarrow$ Positive

5 $\rightarrow$ Negative

$\therefore$ Entropy, (S) =

$$= - \frac{P_{yes}}{P_{Total}} \log_2 \left(\frac{P_{yes}}{P_{Total}} \right) - \frac{P_{No}}{P_{Total}} \log_2 \left(\frac{P_{No}}{P_{Total}} \right)$$

$$= - \frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right)$$

$$= 0.94$$

Step 02

Attribute: Age

Values (Age) = {youth, middle, senior}

$$S_{youth} = [2+, 3-] \rightarrow \text{Total (5)}$$

$$\text{Entropy } (S_{youth}) =$$

$$= - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$= 0.971$$

$$S_{\text{middle}} = [4+, 0-]$$

$$\begin{aligned} \text{Entropy}(S_{\text{middle}}) &= -\frac{4}{4} \log_2\left(\frac{4}{4}\right) - \frac{0}{4} \log_2(0) \\ &= 0 \end{aligned}$$

$$S_{\text{senior}} = [3+, 2-]$$

$$\begin{aligned} \text{Entropy}(S_{\text{senior}}) &= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \\ &= 0.971 \end{aligned}$$

$$\text{Gain}(S, \text{Age})$$

$$= \text{Entropy}(S) - \sum \frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v)$$

$$= 0.94 - \left\{ \left(\frac{5}{14} \times 0.971\right) + \left(\frac{4}{14} \times 0\right) + \left(\frac{5}{14} \times 0.971\right) \right\}$$

$$= 0.2464$$

Continuous

Topic

Linear Regression - slides

Continuous value

x_1	x_2	x_3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

Initial $w_0 = w_1 = w_2 = w_3 = 1$
 $\alpha = 0.05$

Answer :-

Here, $\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$

$\hat{y} = 1 + x_1 + x_2 + x_3$

$[w_0 = w_1 = w_2 = w_3 = 1]$

Slope Bias and weight w_0

$$S_0 = \sum_{i=1}^3 (y(i) - \hat{y}(i)) \cdot x_0(i)$$

x_1	x_2	x_3	y	$\hat{y}$	$y - \hat{y}$
2	7	6	6.5	16	9.5
4	8	7.5	7.8	20.5	12.7
6	9	8.5	8.8	24.5	15.7

Slope for a weight w_0 ,

$$S_0 = \sum_{i=1}^3 (\hat{y}(i) - y(i)) \cdot x_0(i)$$

$$= 9.5 \times 1 + 12.7 \times 1 + 15.7 \times 1$$

$$= 37.9$$

Update weight w_0

$$= w_0 - \alpha \cdot S_0$$

$$= 1 - 0.05 \times 37.9$$

$$= -0.9$$

Slope for a weight w_1 ,

$$S_1 = 9.5 \times 2 + 12.7 \times 4 + 15.7 \times 6$$

$$= 189.4$$

Update weight w_1

$$= 1 - 0.05 \times 189.4$$

$$= -8.47$$

Slope

$$S_2 = 9.5 \times 7 + 12.7 \times 8 + 15.7 \times 9$$

$$= 309.4$$

Update weight w_2

$$= 1 - 0.05 \times 309.4 = -14.47$$

Slope for w_3

$$S_3 = 9.5 \times 6 + 12.7 \times 7.5 + 15.7 \times 8.5 = 285.7$$

Update weight w_3

$$= 1 - 0.05 \times 285.7 = -13.29$$

After 1st iteration,

$$w_0 = -0.9$$

$$w_1 = -8.47$$

$$w_2 = -14.47$$

$$w_3 = -13.29$$

Update equation,

$$\hat{y} = -0.9 - 8.47x_1 - 14.47x_2 - 13.29x_3$$

Update table values,

x_1	x_2	x_3	y	$\hat{y}$	$\hat{y} - y$
2	7	6.3	6.5	-198.87	-205.37
4	8	7.5	7.8	-250.215	-258.015
6	9	8.5	8.8	-294.915	-303.715

Topic - 3

Logistic Regression

Problem 2 iteration of the gradient descent to get update weights w_0, w_1, w_2, w_3 . [Initial $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$]

x_1	x_2	x_3	y
2	-7	6	No
4	-8	7.5	Yes
6	-9	8.5	Yes

Answer :-

Here, $Z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$

$Z = 1 + x_1 + x_2 + x_3$

$\hat{y} = \frac{1}{1 + e^{-Z}}$

x_1	x_2	x_3	y	Z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	2	0.881	0.881
4	-8	7.5	1	4.5	0.989	-0.011
6	-9	8.5	1	6.5	0.998	-0.002

Slope for w_0 ,

$$S_0 = 0.881 * 1 + (-0.011 * 1) + (-0.002 * 1)$$

$$= 0.868$$

Update weight, $w = w - \eta \cdot \text{slope}$

$$w_0 = 1 - 0.05 * 0.868$$

$$= 0.956$$

Slope for w_1 ,

$$S_1 = 0.881 * 2 + (-0.011 * 4) + (-0.002 * 6)$$

$$= 1.706$$

Update weight, $w = w - \eta \cdot \text{slope}$

$$w_1 = 1 - 0.05 * 1.706$$

$$= 0.914$$

Slope for w_2 ,

$$S_2 = 0.881 * (-7) + (-0.011 * (-8)) + (-0.002 * (-9))$$

$$= -6.061$$

Update weight,

$$w_2 = 1 - 0.05 * (-6.061)$$

$$= 1.303$$

slope for w_3 ,

$$S_3 = 0.881 * 6 + (-0.011) * 7.5 + (-0.002) * 8.5$$

$$S_3 = 5.1865$$

Update weight w_3 ,

$$w_3 = 1 - 0.05 * 5.1865$$

$$= 0.74$$

After 1st iteration

$$w_0 = 0.96$$

$$w_1 = 0.91$$

$$w_2 = 1.30$$

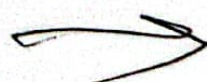
$$w_3 = 0.74$$

update activation function:-

$$z = 0.96 + 0.91x_1 + 1.30x_2 + 0.74x_3$$

Updated table,

x_1	x_2	x_3	y	z	$\hat{y}$	$\hat{y} - y$
2	-7	6	0	-1.88	0.132	0.132
4	-8	7.5	1	-0.25	0.438	-0.562
6	-9	8.5	1	1.01	0.733	-0.267



Q. Draw the ROC curve.

Ave

Curve

$$\frac{TP}{P}$$

$$\frac{FP}{N}$$

Tuple	Class	Prob	TP	FP	TN	FN	TPR (y)	PPR (x)
1	P	0.90	1	0	5	4	1/5 = 0.2	0
2	P	0.80	2	0	5	3	2/5 = 0.4	0
3	N	0.70	2	1	4	3	2/5 = 0.4	0.2
4	P	0.60	3	1	4	2	3/5 = 0.6	0.2
5	P	0.55	4	1	4	1	4/5 = 0.8	0.2
6	N	0.54	4	2	3	1	4/5 = 0.8	0.4
7	N	0.53	4	3	2	1	4/5 = 0.8	0.6
8	N	0.51	4	4	1	1	4/5 = 0.8	0.8
9	P	0.50	5	4	1	0	5/5 = 1.0	0.8
10	N	0.40	5	5	0	0	5/5 = 1.0	1.0

Q

TP = 5, FP = 0, TN = 1, FN = 0

A

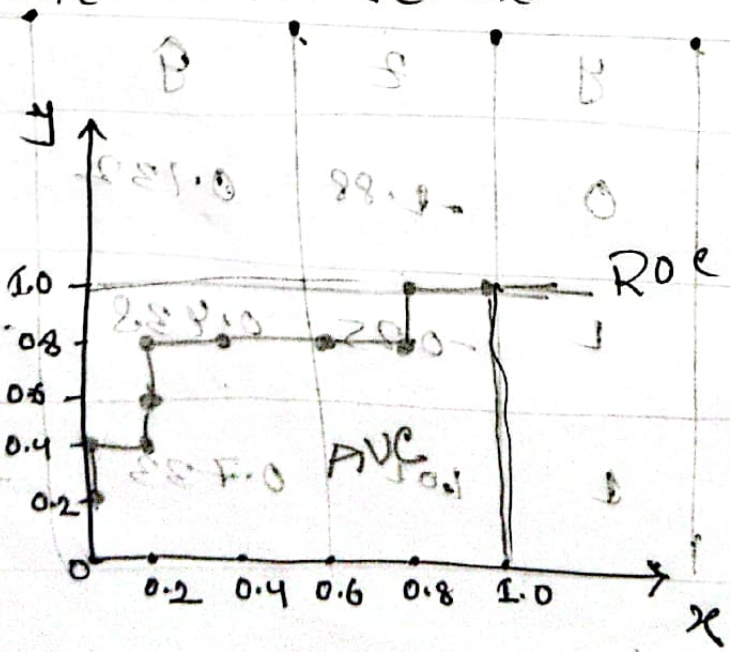
maximizing

সকল

TP → FN এর কমে যাও

FP → FN এর কমে যাও

TP
FP



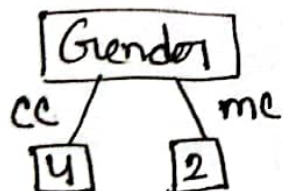
Ada Boost

TE = Total error

Male = no (0)
Female = yes (1)
Bachelor = no (0)
Masters = yes (1)

question

Gender	Age	Education	Income	Get married	Sample Weight	Update Weight	Normalized weight	class Distribution
Male (no)	22 (no)	Bachelor (no)	10 (no)	No	1/6	0.7	0.09	0 = 0.09
Female (yes)	30 (yes)	Masters (yes)	28 (yes)	Yes	1/6	0.7	0.09	0.09 + 0.18
Female (yes)	29 (yes)	Masters (yes)	31 (yes)	Yes	1/6	0.7	0.09	0.18 - 0.27
Male (no)	30 (yes)	Masters (yes)	37 (yes)	Yes	1/6	0.7	0.09	0.27 - 0.36
Male (no)	27 (yes)	Bachelor (no)	20 (no)	No	1/6	0.7	0.09	0.36 - 0.45
Female (yes)	23 (no)	Bachelor (no)	25 (yes)	No	1/6	0.37	0.51	0.45 - 0.96

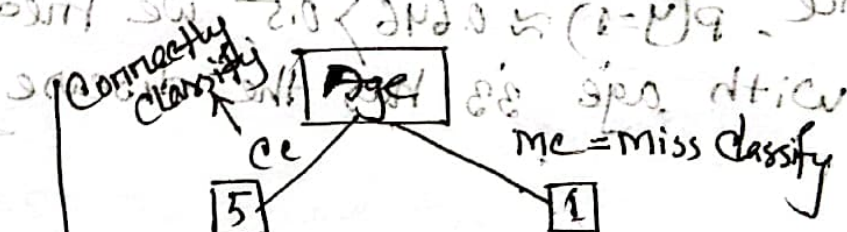


Total error -

Gender miss classification
all (m)

$$= \frac{2}{6} = \frac{1}{3}$$

$$\begin{aligned} \text{Performance say} &= \frac{1}{2} \log \left(\frac{1-TE}{TE} \right) \\ &= \frac{1}{2} \log \left(\frac{1 - \frac{1}{3}}{\frac{1}{3}} \right) \\ &= 0.34 \end{aligned}$$

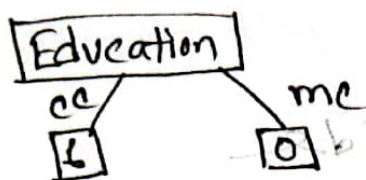


$$\text{Age} = \frac{22+30+29+30+27+23}{6}$$

$$= \frac{201}{6} = 33.5$$

$$\text{Total error} = \frac{1}{6}$$

$$\begin{aligned} \text{Performance say} &= \frac{1}{2} \log \left(\frac{1-TE}{TE} \right) \\ &= \frac{1}{2} \log \frac{1 - \frac{1}{6}}{\frac{1}{6}} \\ &= 0.804 \end{aligned}$$



Total error = 0

Performance Say : 0



$$\text{Income} = \frac{10 + 28 + 31 + 37 + 20 + 25}{6}$$

$$= 22.16 \approx 23$$

Performance Say: $\frac{1}{2} \log \left(\frac{1 - \text{TF}}{\text{TF}} \right)$

$$= \frac{1}{2} \log \frac{1 - (1/6)}{1/6}$$

$$= 0.804$$

New weight for misclassify = Performance

Sample weight * $e^{\text{PS}(\max)}$

$$= 1/6 * e^{0.804}$$

$$= 0.37$$

and, new weight for classify = 1 - new weight for misclassify

Sample weight * $e^{-\text{PS}(\max)}$

$$= 1/6 * e^{-0.804}$$

$$= 0.07$$

Normalized weight = $0.7 + 0.7 + 0.7 + 0.7 + 0.7 + 0.37 = 0.72$

$$\frac{0.7}{0.72} = 0.97$$

$$\frac{0.37}{0.72} = 0.51$$

	cc	mc		cc	mc
1	1	0	5	1	1
2	1	0	28	1	1
3	1	0	31	1	1
4	1	0	37	1	1
5	1	0	20	1	1
6	1	0	25	1	1

Linear Regression (7)

⑧

Temperature (°C)	Ice-Cream Sales
25	200
30	250
20	180
28	220
35	330

Predict the ice cream sales if the temperature is 32°C

⇒

x Temperature	y ice-cream sales	Mean of (x)	Mean of (y)	Deviation of x (x - $\bar{x}$)	Deviation of y (y - $\bar{y}$)	Product of (x-y)	Squared deviation of (x)
25	200	27.6	236	25 - 27.6 = -2.6	200 - 236 = -36	(-2.6) × (-36) = 93.6	(-2.6) ² = 6.76
30	250			30 - 27.6 = 2.4	250 - 236 = 14	2.4 × 14 = 33.6	5.76
20	180			20 - 27.6 = -7.6	180 - 236 = -56	-7.6 × -56 = 425.6	57.76
28	220			28 - 27.6 = 0.4	220 - 236 = -16	0.4 × -16 = -6.4	0.16
35	330			35 - 27.6 = 7.4	330 - 236 = 96	7.4 × 96 = 705.6	54.76
						Σ 1242	Σ 125.2

$$\text{mean of } (x) = \frac{25 + 30 + 20 + 28 + 35}{5} = 27.6$$

$$\text{mean of } (y) = \frac{200 + 250 + 180 + 220 + 330}{5} = 236$$

Logistic Regression

We know that,

$$y = mx + c$$

Here, $m = \frac{\text{Sum of Product deviation}}{\text{Sum of Square deviation of } x}$

$$= \frac{1242}{125.2}$$

$$= 9.92$$

Now, $c = y - mx$

$$= \{ \text{mean}(y) \} - m \{ \text{mean}(x) \}$$

$$= 236 - (9.92 \times 27.6)$$

$$= -37.79$$

Temp (x)	Disease (y)
22	0
28	0
32	0
40	1
42	1

Now, temperature at 32°C
 $\therefore x = 32^\circ\text{C}$

$$y = mx + c$$

$$= 9.92 \times 32 + (-37.79)$$

$$= 279.65 \approx 280$$

So, Predicted ice-Cream sales will be 280 if the temperature at 32°C

Ans

Decision Tree - With Gini

☞ Gini Sort Outlook

<u>Outlook</u>	<u>Yes</u>	<u>No</u>	<u>Total</u>
Sunny	2	3	5
Rainy	3	2	5
Overcast	4	0	4

Sunny $\Rightarrow$

$$\begin{aligned} \text{Gini} &= 1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \\ &= 0.48 \end{aligned}$$

Rainy $\Rightarrow$

$$\begin{aligned} \text{Gini} &= 1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \\ &= 0.48 \end{aligned}$$

Overcast $\Rightarrow$

$$\begin{aligned} \text{Gini} &= 1 - \left(\frac{4}{4} \right)^2 - \left(\frac{0}{4} \right)^2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Gini (Outlook)} &= \left(\frac{5}{14} \right) * 0.48 + \left(\frac{5}{14} \right) * 0.48 + \\ &\quad \left(\frac{4}{14} \right) * 0 \\ &= 0.342 \end{aligned}$$

Q Gini For Temp.

<u>Temp</u>	<u>Yes</u>	<u>No</u>	<u>Total</u>
Hot	2	2	4
Mild	4	2	6
Cold	3	1	4

Hot $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2$$

$$= 0.5$$

Mild $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{4}{6}\right)^2 - \left(\frac{2}{6}\right)^2$$

$$= 0.445$$

Cold $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2$$

$$= 0.375$$

$$\text{Gini (Temp.)} = \left(\frac{4}{14}\right) * 0.5 + \left(\frac{6}{14}\right) * 0.44$$

$$+ \left(\frac{4}{14}\right) * 0.375$$

$$= 0.439$$

Q Gini For Humidity

<u>Humidity</u>	<u>Yes</u>	<u>No</u>	<u>Total</u>
High	3	4	7
Normal	6	1	7

High $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2$$

$$= 0.489$$

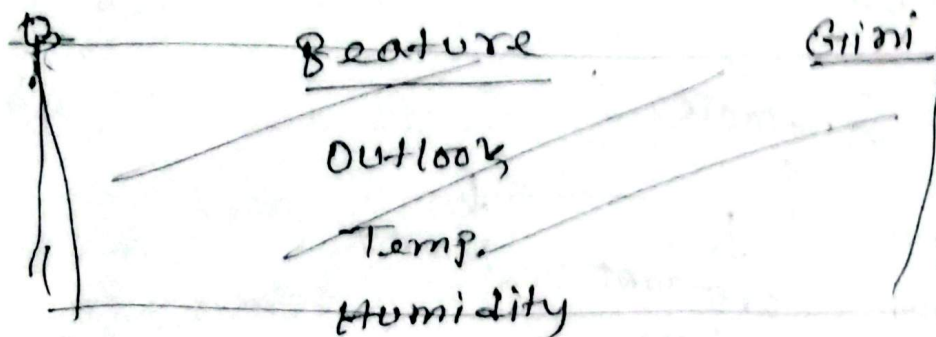
Normal $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2$$

$$= 0.244$$

$$\text{Gini (Humidity)} = \left(\frac{7}{14}\right) * 0.489 + \left(\frac{7}{14}\right) * 0.244$$

$$= 0.367$$



B Gini for Wind

<u>Wind</u>	<u>yes</u>	<u>no</u>	<u>Total</u>
Weak	6	2	8
Strong	3	3	6

Weak $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{6}{8}\right)^2 - \left(\frac{2}{8}\right)^2$$

$$= 0.375$$

Strong $\Rightarrow$

$$\text{Gini} = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2$$

$$= 0.5$$

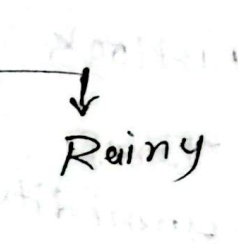
$$\text{Gini (Wind)} = \left(\frac{8}{14}\right) * 0.375 + \left(\frac{6}{14}\right) * 0.5$$

Pini

Humidity 0.367
Wind 0.428
Smallest Gini = 0.171

(4) *Simulium* ?

≈ 0.671605



<u>Temp.</u>	<u>Humidity</u>	<u>wind</u>	<u>Decision</u>
Hot	High	Weak	No
Hot	High	Strong	No
Mild	High	Weak	No
Cool	Normal	Weak	Yes
Mild	Normal	Strong	Yes

cc - Mr. J. B. ...

Scenario: Credit Card Fraud Detection

A bank builds a machine learning model to detect fraudulent transactions.

Out of **5000 transactions**:

- 300 transactions are actually fraudulent
- The model predicts **400 transactions as fraud**
- Among them, **240 are actually fraud**

Additionally, for predicting **transaction amount error (regression task)**:

Transaction	Actual Amount (\$)	Predicted Amount (\$)
1	100	90
2	250	270
3	400	380
4	150	180

For classification probability:

- True label = 0 (not fraud)
- Predicted probability of fraud = 0.3

Evaluation Matrices and Loss Function

1. Find out TP, FP, FN, TN
2. Calculate Precision, Recall, F1-score.
3. Calculate Regression Loss MAE and MSE
4. Calculate Binary cross-entropy loss.
5. calculate. R² error and MCE

1

$$\text{True Positive (TP)} = 240$$

$$\text{False positive (FP)} = 400 - 240 = 160$$

$$\text{False Negative (FN)} = 300 - 240 = 60$$

$$\begin{aligned}\text{True Negative (TN)} &= 5000 - (240 + 160 + 60) \\ &= 4560\end{aligned}$$

2 Metrics.

$$\text{Sensitivity or Precision: } \frac{TP}{TP + FP} = \frac{240}{240 + 160} = 0.6$$

$$\text{Recall: } \frac{TP}{TP + FN} = \frac{240}{240 + 60} = 0.8$$

$$\begin{aligned}\text{F1-Score: } 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ = 2 \times \frac{0.6 \times 0.8}{0.6 + 0.8} = 0.688\end{aligned}$$

Q. Loss function, Regression.

$$\text{Error} = |\text{Actual} - \text{predicted}|$$

Transaction

error

1

$$100 - 90 = 10$$

2

$$250 - 270 = 20$$

3

$$400 - 380 = 20$$

4

$$150 - 180 = 30$$

$$\text{MAE} = \frac{10 + 20 + 20 + 30}{4} = 20$$

$$\text{MSE} = \frac{(10)^2 + (20)^2 + (20)^2 + (30)^2}{4} = 450$$

y

$$y = 0$$

$$p = 0.3$$

$$\begin{aligned} \text{Entropy} &= -[y \log(p) + (1-y) \log(1-p)] \\ &= -[0 \cdot \log(0.3) + (1-0) \cdot \log(1-0.3)] \\ &= 0.357 \end{aligned}$$

15

$$R^2 = 1 - \frac{\sum (y - \hat{y})^2 \rightarrow \text{error}}{\sum (y - \bar{y})^2}$$

$$\text{mean } \bar{y} = \frac{100 + 250 + 400 + 150}{4} = 225$$

$$R^2 = 1 - \frac{(10)^2 + (20)^2 + (20)^2 + (30)^2}{(100 - 225)^2 + (250 - 225)^2 + (400 - 225)^2 + (150 - 225)^2}$$

$$= 1 - \frac{1800}{52500}$$

$$= 0.966$$

$$TP \cdot TN - FP \cdot FN$$

$$\Rightarrow MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$= \frac{240 \cdot 4540 - 160 \cdot 60}{\sqrt{(240 + 160)(240 + 60)(4540 + 160)(4540 + 60)}}$$

$$= 0.67$$

⊕ Analytical Question

- In this fraud detection system, which is more important: precision or Recall? Why?
- If the bank wants to reduce customer annoyance, which metric should be prioritized?
- If fraud cases are very rare (Imbalanced dataset), why is accuracy a bad metric here?
- When should we prefer F1-score over precision and Recall individually?
- Why is cross-entropy loss preferred over MSE in classification problems?